

Learner Name

Mr. Abdi Farah

1. Did you use any AI tool(s) to assist with this assignment?

☐ No. Why not?

☒ Yes. List the AI tool(s) you used and briefly describe the purpose for which you used the tool(s). (e.g., brainstorming, research, drafting content)

I used ChatGPT as an assistance tool. The use ChatGPT brainstorming the report structure, organizing the assignment contents, drafting and refining explanations of ANN optimization, and checking the completeness of the submission package. The AI was not used in place of doing execution of the models and personal review. The dataset was processed, the ANN variants were trained, and final metrics were analyzed within the context of the marketing segmentation goal.

2. To demonstrate effective AI use, please provide your initial prompt(s) and a brief description of the steps you took to refine and iterate on the AI's output for this

My initial prompts concentrated on refining the optimization of Artificial Neural Networks (ANNs) for the purpose of market segmentation, including the framework for algorithm selection and performance transparency in relation to the marking rubric. The segmentation Artificial Intelligence (AI) output was assessed in relation to the segmentation task specification, which included the reconstruction of the initial model, the application of no less than three optimization algorithms, the evaluation of performance metrics, and a justification of the role of the optimized model of segmentation in aiding Otomoto's customer segmentation and marketing efforts. Segmentation task specification also included delineation of additional metrics such as accuracy, precision, recall, F1-score, and ROC-AUC, as well as a more comprehensive description of the segmentation model, chosen optimizers, and their effect on segmentation. The framings of AI used to respond to the segmentation task were overly broad and required a task-based interpretation. AI also d

3. List at least one source you consulted to verify the AI-generated content in your work.

AI-generated outputs were checked and validated against the following sources:
Chicco, D., & Jurman, G. (2020). The advantages of the Matthews correlation coefficient over F1 score and accuracy in binary classification evaluation. BMC Genomics, 21, 6. Kingma, D. P., & Ba, J. (2015). Adam: A method for stochastic optimization. International Conference on Learning Repr